

GE261004 DESIGN THINKING

Unit V – Innovation, Entrepreneurship and Sustainability · Presentation 3 of 6

Sustainable and Inclusive Design

Counting the whole life · linear and circular models · design decisions that change the life cycle · universal design · engineering and the SDGs

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In this presentation

- What sustainable design means
- The whole life of a product
- Linear and circular
- Decisions that change the life cycle
- The seven principles
- Designing for the edges
- Four exclusions engineers forget
- Engineering and the SDGs

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Designing for the whole life

The factory is a small part of the footprint

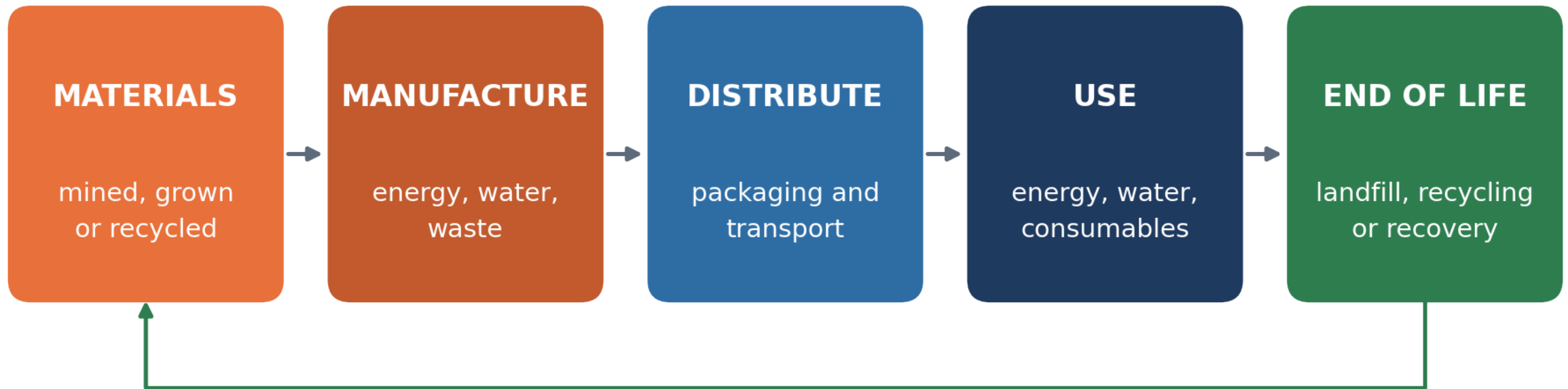
What sustainable design means

- Meeting the need now without making it harder to meet the same need later
- It is a set of design decisions, not a material choice made at the end
- Most of a product's impact is fixed in the first few weeks of designing it
- It is also usually cheaper – less material, less energy, fewer parts, longer life

KEY POINT By the time you are choosing between two plastics, most of the footprint has already been decided.

The whole life of a product

Count the whole life, not the factory



for anything that consumes energy, the use phase usually dominates

Life-cycle thinking: count materials, making, moving, using and disposing.

Linear and circular

LINEAR

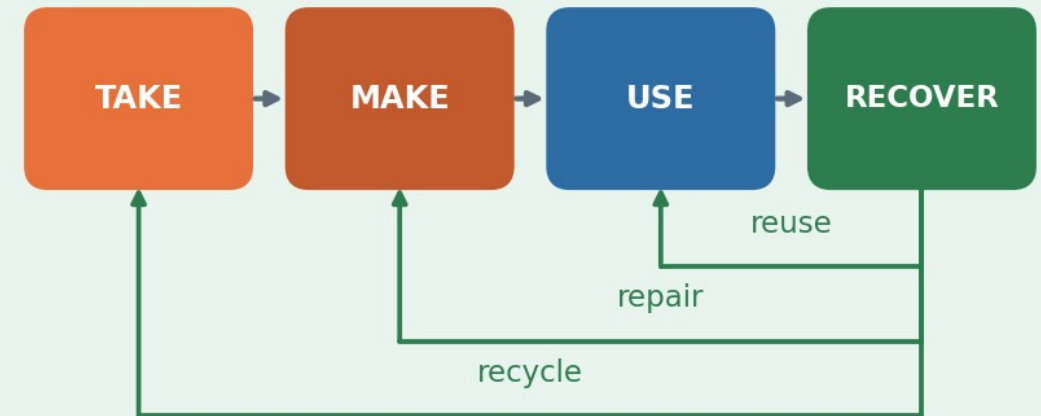
take – make – dispose



value ends in a bin

CIRCULAR

keep it in use, at value



Popularised by the Ellen MacArthur Foundation: keep materials in use.

Decisions that change the life cycle

- **Fewer parts** – less material, less assembly, fewer things to fail
- **Repairable** – screws not glue, standard fasteners, a spare you can actually buy
- **Efficient in use** – for anything that consumes energy, this usually dominates everything else
- **Separable at end of life** – materials that can be told apart can be recovered

KEY POINT Design for disassembly. If nobody can take it apart, nothing in it will ever be recovered.

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Inclusive design

Who your design quietly leaves out

The seven principles of universal design

1

Equitable use

the same way, for everyone

2

Flexibility in use

more than one way to do it

3

Simple and intuitive

obvious without instruction

4

Perceptible information

seen, heard or felt

5

Tolerance for error

a mistake is not a disaster

6

Low physical effort

usable when tired or weak

7

Size and space

reachable seated or standing

Ronald Mace and the Center for Universal Design, North Carolina State University.

Designing for the edges

- Design for the middle and the edges are excluded; design for the edges and the middle follows
- Ability is not fixed – it changes with age, injury, tiredness, noise and light
- Many exclusions are temporary or situational: a broken arm, a crowded bus, bright sun
- Inclusive design is ordinary good design that somebody bothered to check

KEY POINT A ramp built for a wheelchair carries every trolley, pram and delivery cart on the campus.

Four exclusions engineers forget

Cost

The commonest exclusion of all. If they cannot afford it, nothing else about it matters.

Literacy and language

Instructions, labels and error messages in one language, at one reading level.

Physical ability

Grip strength, reach, sight, hearing, balance – and how all of these change with age.

Connectivity and power

Assuming a smartphone, a data plan and mains electricity that is always there.

Engineering and the SDGs

- The seventeen Sustainable Development Goals were adopted in 2015, with targets to 2030
- Engineering work sits directly under several: water, energy, health, cities, industry and consumption
- Use them to check who your project helps and who it might harm, not as decoration on a slide
- A single honest sentence about one goal beats a wall of logos

KEY POINT Name one goal your project genuinely serves and one it could work against. Both answers are useful.

In class: audit your own concept

- Walk your concept through the five life-cycle stages and name the biggest impact
- Check it against the seven principles and find the one it fails worst
- Name one person your design currently excludes
- Write one change that would fix that exclusion

ACTIVITY Twenty-five minutes. Each team reports its biggest impact, its worst principle and its one change.

THINK ABOUT IT

Who cannot use what your team designed – and did you notice before this slide?

- Is the exclusion about cost, ability, language or infrastructure?
- How many people does it affect?
- What would it cost to include them?
- What do you lose by including them – honestly?

Summary

- Sustainability is decided in the first weeks of design, not in material choice
- Count the whole life: materials, making, moving, using, disposing
- Circular thinking keeps materials in use through reuse, repair and recycling
- Seven principles of universal design, from Mace and NC State
- Cost, literacy, physical ability and connectivity are the forgotten exclusions
- Use the SDGs to check who you help and who you might harm

Next: Social innovation and the product development cycle

- What makes social innovation sustainable
- Cross-subsidy, frugal engineering, community ownership
- Stage-Gate from concept to launch
- What actually happens at a gate